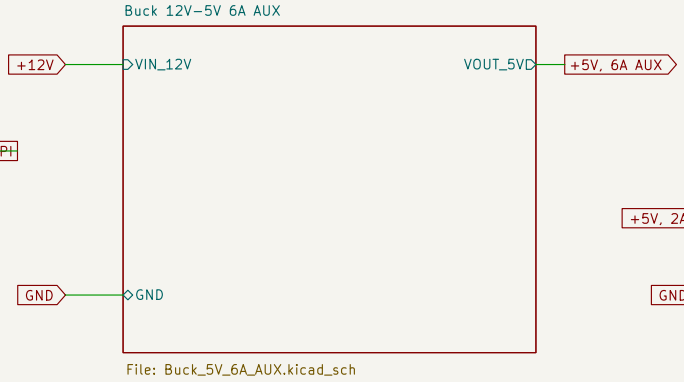
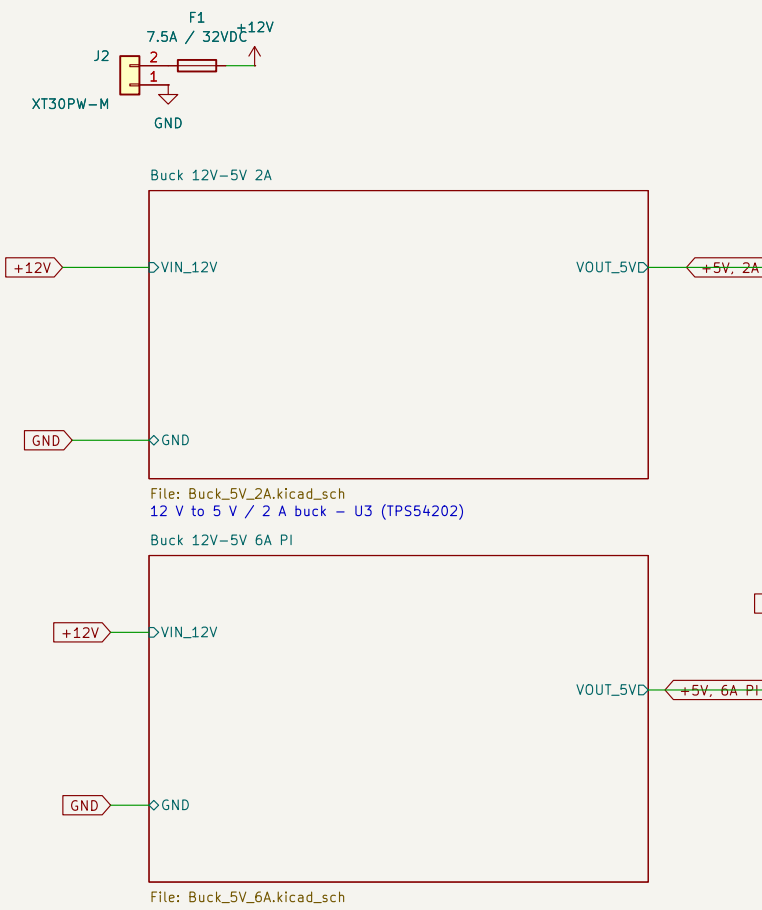
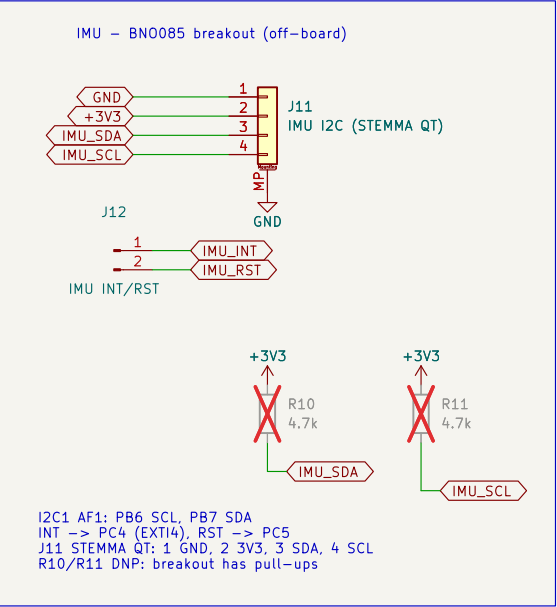
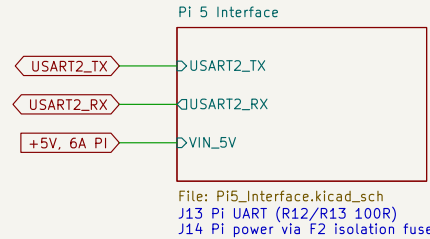
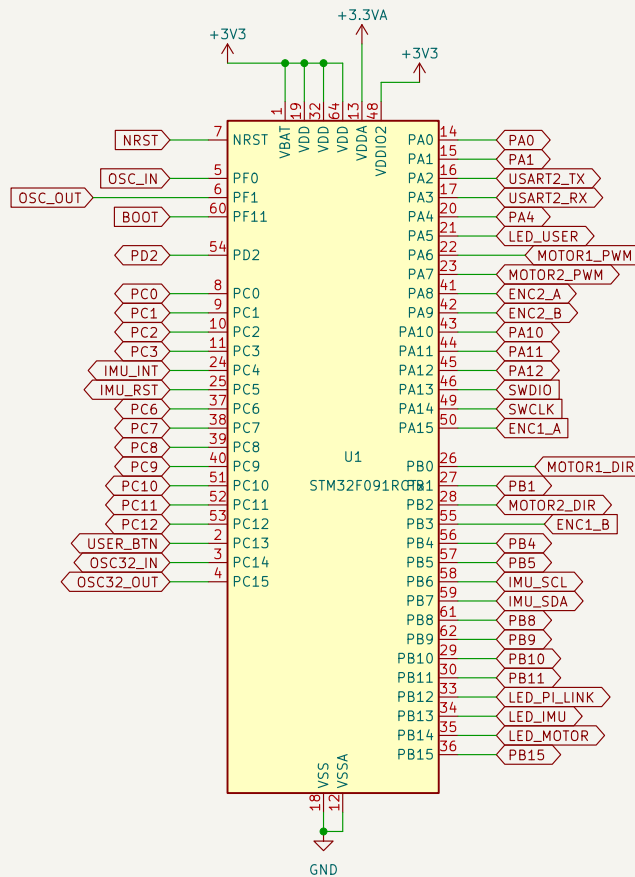
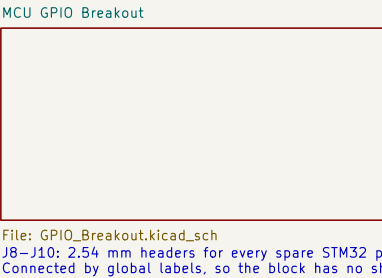
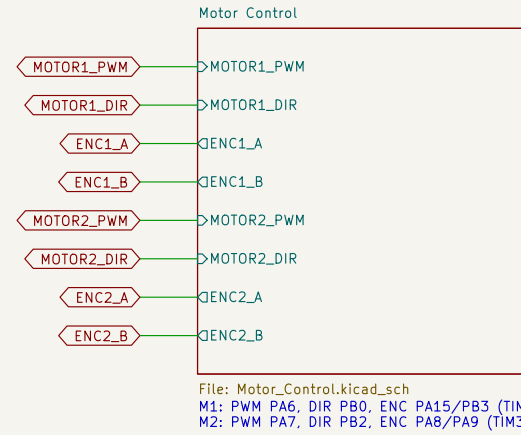
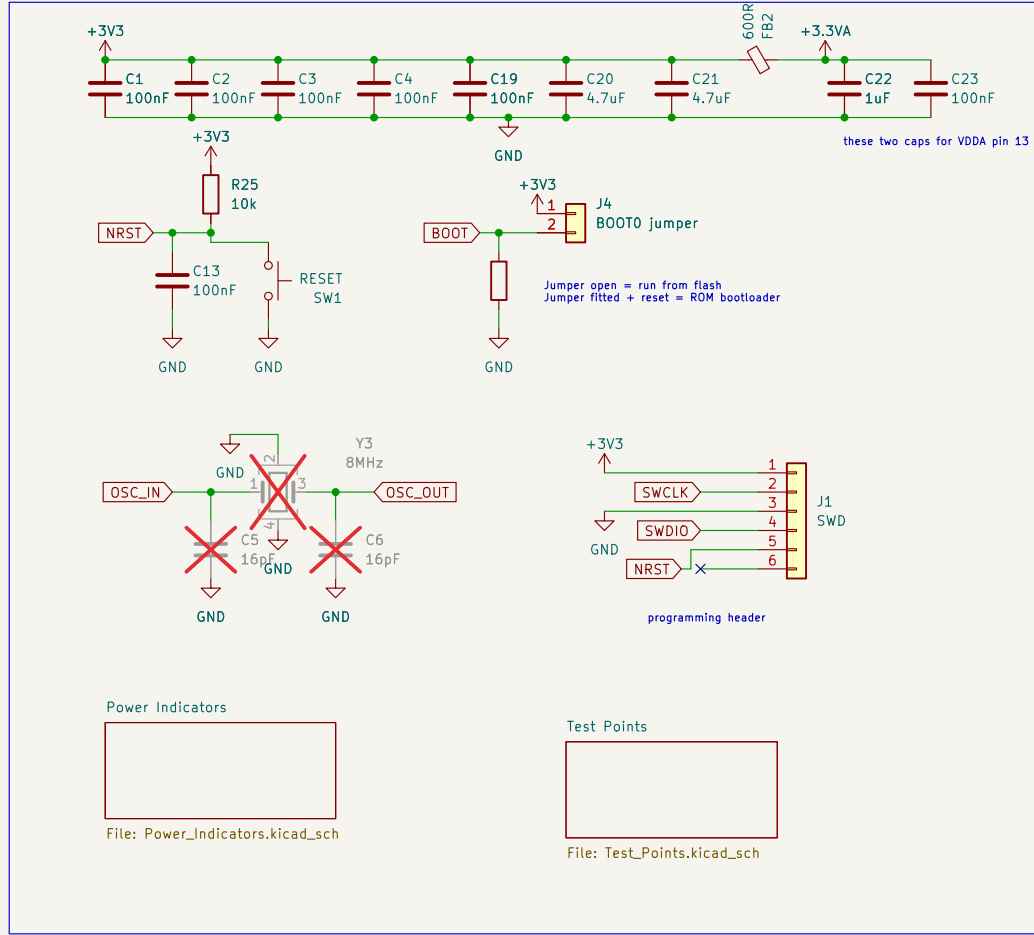
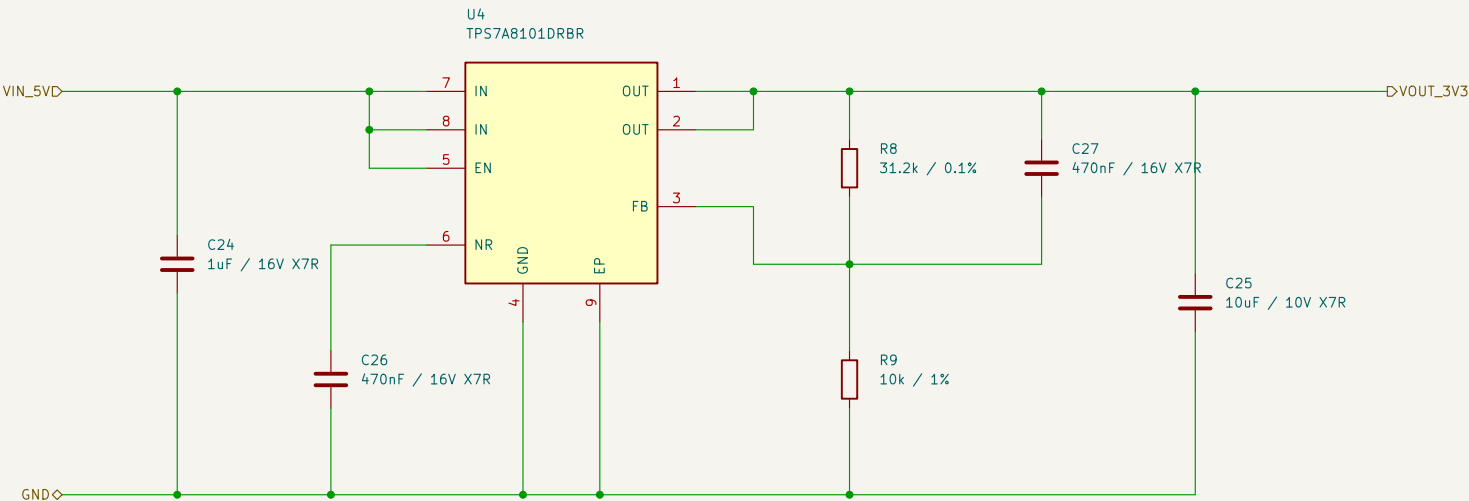


3.3 V / 1 A LDO - U4
STM32 VDD and analog filter input supplied from +3V3

5 V to 3.3 V LDO supply

Reusable sheet: VIN_5V input, VOUT_3V3 output, common GND



VOUT nominal = $0.8 \times (1 + 31.2k / 10k) = 3.296 \text{ V}$
1 A is the TOTAL output budget for STM32 + IMU + other 3.3 V loads.

C2: require $\geq 4.7\mu\text{F}$ EFFECTIVE capacitance; selected part modeled at 8.8uF.
EN tied to IN. C3 sets noise reduction / soft start; C4 bypasses R1.

Layout: capacitors close to U1; EP pad 9 to GND copper and thermal vias.
Heat at 1 A: about 1.7 W at 5 V input; verify junction temperature on final PCB.

5 V motor / encoder supply branches BEFORE this sheet.
Adafruit BNO085 breakout: 3.3 V rail to VIN; 3Vo is an output, leave separate.
Bare BNO085: connect 3.3 V to VDD and VDDIO with its specified decoupling.

5 V to 3.3 V, 1 A total rail capacity

Senior Design

Sheet: /LDO Power/
File: LDO_3V3.kicad_sch

Title: LDO power supply – reusable sheet

Size: A4

Date: 2026–10–05

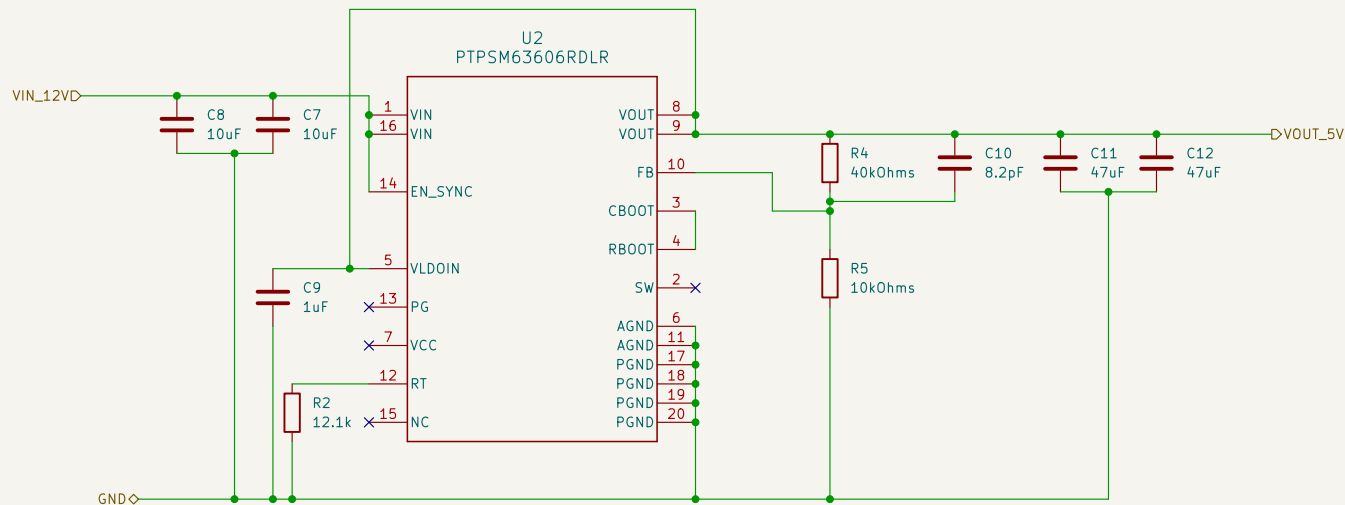
Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 2/10

12 V – 5 V, 6A Buck Convertor

12 V to 5 V buck converter – TPSM63606 (U2)



Reusable sheet: VIN_12V input, VOUT_5V output, common GND
VOUT set by R4/R5 divider on FB; RT = R2; EN tied to VIN.
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

12 V to 5 V buck, TPSM63606 (U2), 6 A rail

Senior Design

Sheet: /Buck 12V-5V 6A PI/

File: Buck_5V_6A.kicad_sch

Title: Buck converter – reusable sheet

Size: A4

Date: 2026-10-06

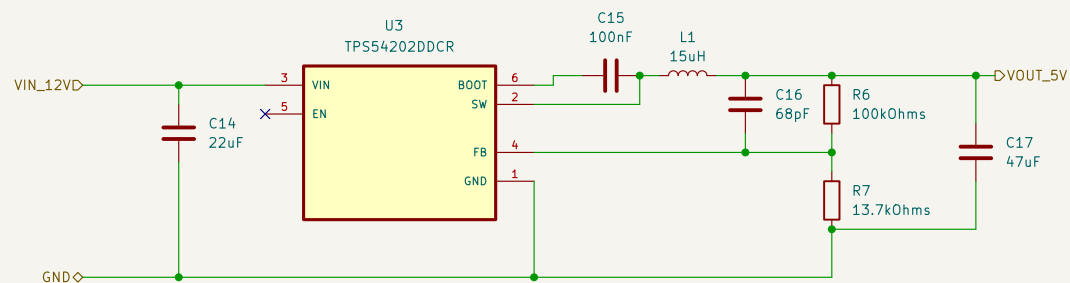
Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 3/10

12 V – 5 V, 2A Buck Convertor

12 V to 5 V buck converter – TPS54202 (U3)



Reusable sheet: VIN_12V input, VOUT_5V output, common GND

12 V to 5 V buck, TPS54202 (U3), 2 A rail
Senior Design

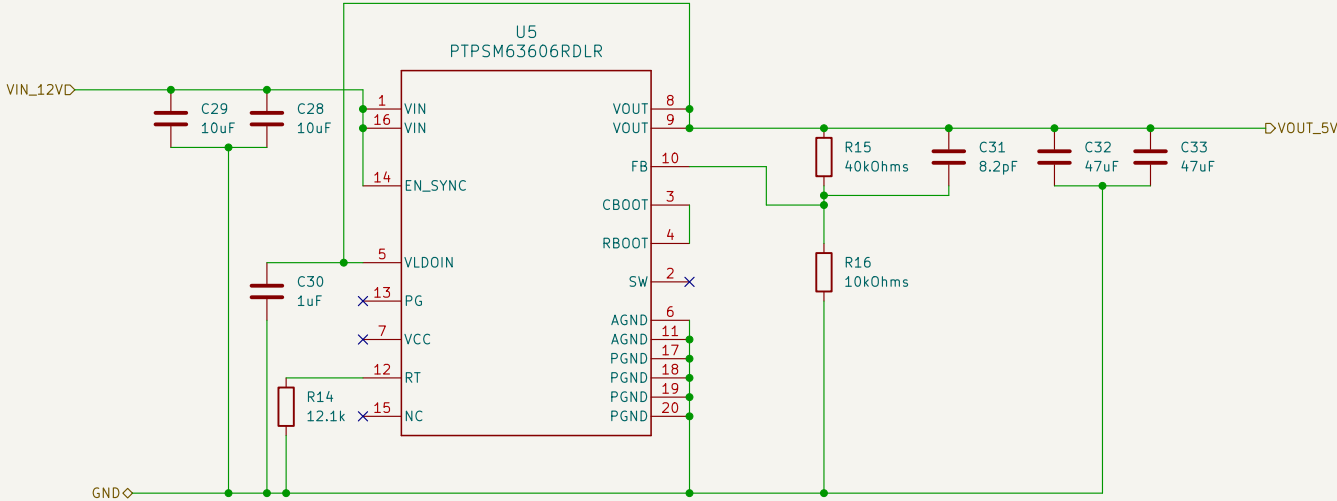
Sheet: /Buck 12V–5V 2A/
File: Buck_5V_2A.kicad_sch

Title: Buck converter – reusable sheet

Size: A4	Date: 2026–10–06	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 4/10

12 V – 5 V, 6A Buck Convertor

12 V to 5 V buck converter – TPSM63606 (U5)



Reusable sheet: VIN_12V input, VOUT_5V output, common GND.
VOUT set by R15/R16 divider on FB; RT = R14; EN tied to VIN.
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

12 V to 5 V buck, TPSM63606 (U5), 6 A rail

Senior Design

Sheet: /Buck 12V-5V 6A AUX/

File: Buck_5V_6A_AUX.kicad_sch

Title: Buck converter – reusable sheet

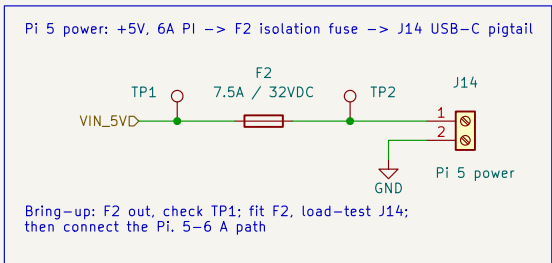
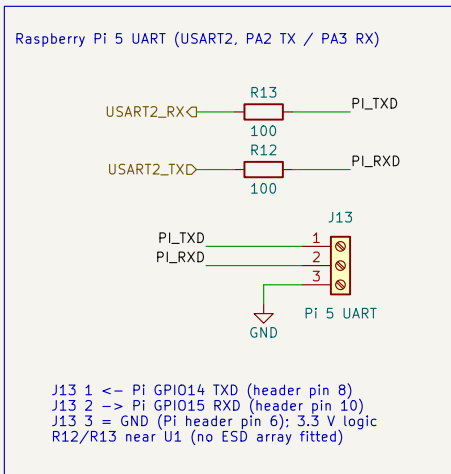
Size: A4

Date: 2026-10-06

Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 5/10

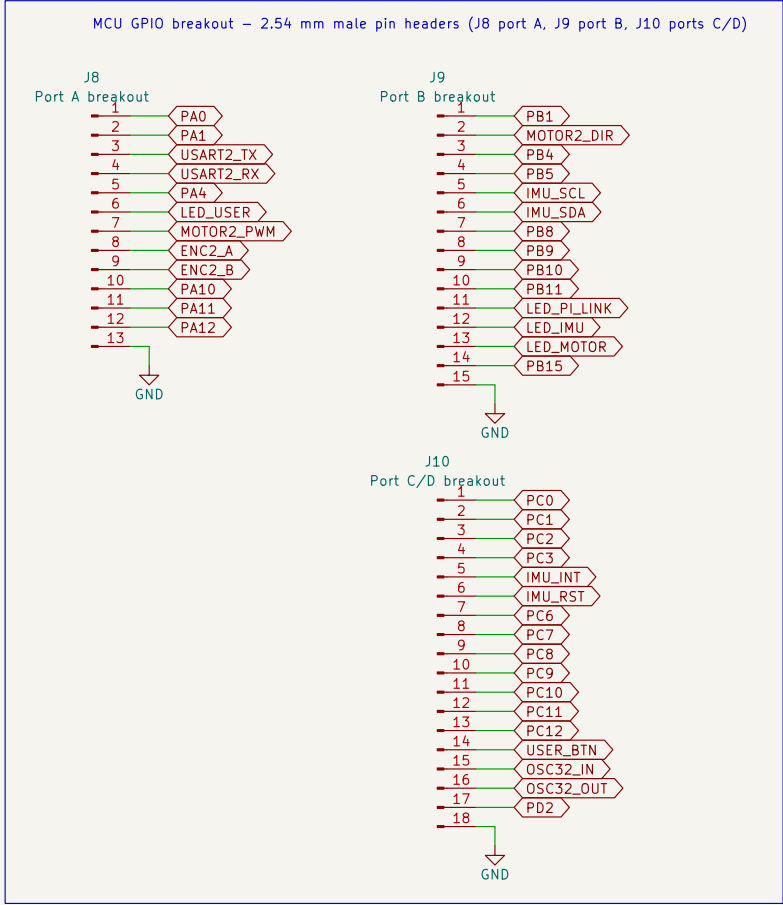


Senior Design

Sheet: /Pi 5 Interface/
File: Pi5_Interface.kicad_sch

Title: Raspberry Pi 5 interface – UART and power

Size: A4	Date: 2026-10-07	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 6/10



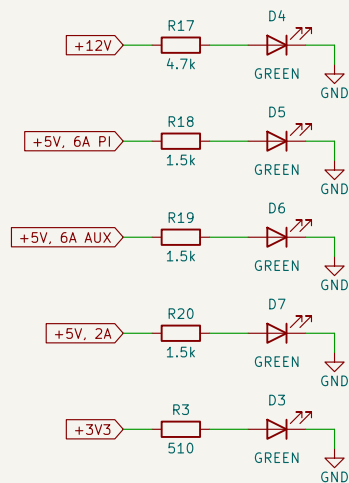
Senior Design

Sheet: /MCU GPIO Breakout/
File: GPIO_Breakout.kicad_sch

Title: STM32 GPIO breakout headers

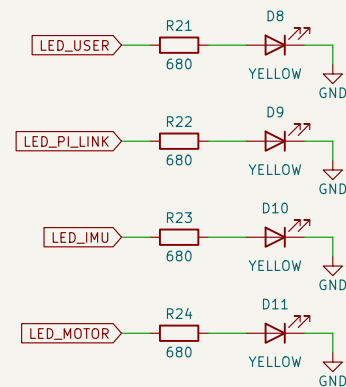
Size: A4	Date: 2026-10-07	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 7/10

Power rail indicators – one green LED per rail
(Lite-On LTST-C190KGKT, 0603, V_f 2.0 V typ)



LED current ~2–2.7 mA: $R = (V_{rail} - 2.0 \text{ V}) / I$. +12V uses 4.7k for the 10.8–14.6 V battery range.
All resistors 0805; worst-case dissipation 34 mW (R17 at 14.6 V).

Debug LEDs – STM32 GPIO, active high (Lite-On LTST-C190KSKT yellow, 0603)



PA5 LED_USER: heartbeat
PB12 LED_PI_LINK: PI UART activity
PB13 LED_IMU: IMU data-ready
PB14 LED_MOTOR: motor running / fault
680R: $(3.3 - 2.1) / 680 = 1.8 \text{ mA per pin}$

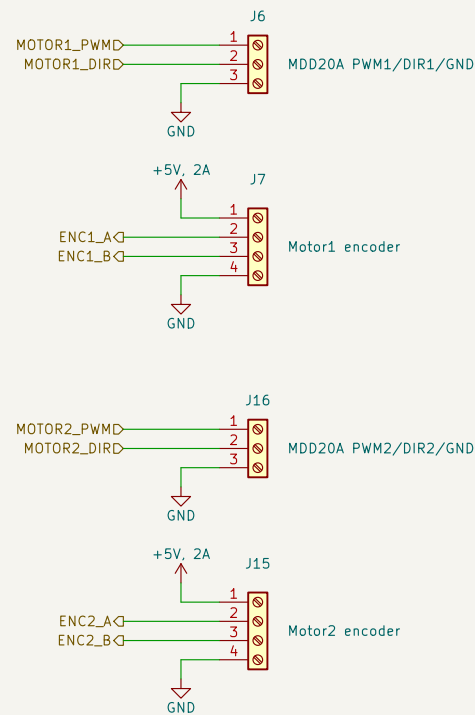
Senior Design

Sheet: /Power Indicators/
File: Power_Indicators.kicad_sch

Title: Power rail and debug indicator LEDs

Size: A4	Date: 2026-10-07	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 8/10

Motor control – Cytron MDD20A dual-channel driver + two 37D encoder motors



STM32F091 PIN ASSIGNMENT

Motor 1 (J6 driver, J7 encoder)

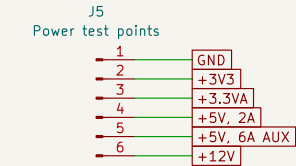
MOTOR1_PWM	PA6	TIM3_CH1	-> MDD20A PWM1
MOTOR1_DIR	PB0	GPIO	-> MDD20A DIR1
ENC1_A	PA15	TIM2_CH1	<- encoder A (yellow)
ENC1_B	PB3	TIM2_CH2	<- encoder B (white)

Motor 2 (J16 driver, J15 encoder)

MOTOR2_PWM	PA7	TIM3_CH2	-> MDD20A PWM2
MOTOR2_DIR	PB2	GPIO	-> MDD20A DIR2
ENC2_A	PA8	TIM1_CH1	<- encoder A (yellow)
ENC2_B	PA9	TIM1_CH2	<- encoder B (white)

Timers: TIM3 = both PWMs (same frequency, <= 20 kHz for MDD20A)
TIM2 = encoder 1, TIM1 = encoder 2 (encoder mode)
Encoders powered from +5V, 2A; A/B pins must be 5 V tolerant (FT).
Driver VB+/VB- (battery) and M1A/M1B, M2A/M2B wire directly to the MDD20A.

Power test points



J5: one 2.54 mm header for every rail, ordered by voltage
1 GND
2 +3V3
3 +3.3VA (MCU analog)
4 +5V, 2A
5 +5V, 6A AUX
6 +12V (after F1)



TP3: GND loop for the scope ground clip.
+5V, 6A PI is on TP1 (PI 5 Interface sheet),
used in the PI bring-up check.

Senior Design

Sheet: /Test Points/
File: Test_Points.kicad_sch

Title: Power test points

Size: A4	Date: 2026-10-08	Rev: 1.1
KiCad E.D.A. 10.0.6	Id: 10/10	